# Big Data Analytics Using R

M20CA3020 · Unit 1: Introduction to Big Data and Analytics · Exam Prep Notes (easy language)
Based on Unit_1 slides (Prof. Nagaraju S). Parts marked "syllabus" are short additions for topics the slides do not explain.

> **Unit 1 at a glance:** Digital data → Types of data → What is Big Data & 5 Vs → Challenges → Big Data vs BI → Architecture (10 layers) → Hadoop tools → 4 Types of Analytics → Analytics Life Cycle (6 phases).

## 1. Digital Data

- **Digital data** = information stored as **binary (0s and 1s)** so that machines/computers can read it.
- It can be text, photos, videos, e-books, etc., shared using electronic devices.

## 2. Types of Data

Data is of 3 types based on its format: **Structured, Unstructured, Semi-structured.**

### a) Structured data

- Highly organized, stored in a **fixed format** (rows and columns) with a predefined **schema**.
- Stored in **relational databases (SQL)**. Easy to search and process.
- Comes from enterprise systems like **ERP and CRM**.
- **Examples:** SQL tables, Excel sheets, CSV files.

### b) Unstructured data

- No predefined structure or schema. Hard to store in normal relational databases.
- Stored in **NoSQL databases**, cloud storage or data lakes. Cannot be queried directly with SQL.
- **Examples:** emails, chat messages, social media photos/videos, X-rays/MRI, IoT sensor data.
- *NoSQL = "Not only SQL": a non-relational database.*

### c) Semi-structured data

- Has **some structure** (tags/keys) but is **not relational**. It is hierarchical or graph-based.
- Stored in text files; easier to process than unstructured data.
- **Examples:** XML, JSON, RDF, emails (metadata + text), MongoDB data.

### Difference table (very important)

| Point | Structured | Semi-structured | Unstructured |
|---|---|---|---|
| Structure | Highly organized, fixed format | Some structure & consistency | No predefined structure |
| Model | Follows data model/schema | Hierarchical or graph-based | No data model/schema |
| Made of | Tables, rows, columns | Key-value pairs, tags, trees | Binary, character, text, media files |
| Based on | Relational database | XML / RDF | Character and binary data |
| Examples | SQL databases, Excel, CSV | JSON, XML, email, NoSQL (MongoDB) | Word/PDF, images, audio, video, social posts |

## 3. What is Big Data?

- **Big Data** = very large and complex collections of data that are **difficult to manage** with traditional tools. It includes structured, semi-structured and unstructured data.
- It is a field about the **analysis, processing and storage** of large data that comes from many different sources.
- It is analysed by computers to find **patterns, trends and associations**, especially about human behaviour.
- "Big" is relative: in 1999, **1 GB** was big data. Today we talk in **petabytes** (1,024 TB) or **exabytes** (1,024 PB), with billions of records.

### Where does Big Data come from? (sources)

- **Social media:** Facebook, WhatsApp, YouTube, Instagram, etc. (billions of users).
- **E-commerce:** Amazon, Flipkart – logs used to find buying trends.
- **Weather stations/satellites:** data used to forecast weather.
- **Telecom companies:** Airtel, Vodafone – study user trends to design plans.
- **Share market:** huge daily transaction data.

> **Quick facts (for examples in answers):** ~6,000 tweets per second (500 million/day); 350 million photos uploaded to Facebook daily; Walmart handles 1 million+ transactions every hour.

### Issues and Solution

- **Issue:** data is of mixed types (structured, unstructured, semi-structured), huge in amount, and hard for normal systems to store, process and analyse.
- **Solution:**
  - **Storage:** Hadoop with **HDFS** – uses cheap *commodity hardware* in clusters; **write once, read many times**. NoSQL (HBase, Cassandra, MongoDB) runs on top.
  - **Processing: MapReduce**.
  - **Analysis: Pig, Hive**.
  - **Cost:** Hadoop is **open source**, so cost is low.

### Evolution of Big Data (in syllabus; short summary)

- Earlier, data was small and kept in files and then relational databases/data warehouses (structured only).
- With the internet, social media, mobiles and IoT, data became huge, fast and varied, so new tools like Hadoop and NoSQL appeared to handle it.

## 4. Characteristics of Big Data: the 5 Vs

A dataset is called Big Data if it has one or more of these: **Volume, Velocity, Variety, Veracity, Value.**

| V | Meaning (easy) | Example / key point |
|---|---|---|
| **Volume** | The **amount** of data generated every second. Grows very fast. | IoT devices, social media, online banking, scientific experiments, customer logs |
| **Velocity** | The **speed** at which data is created and must be processed. Large data piles up in a short time. | Tweets, videos, emails, GBs from a jet engine every minute |
| **Variety** | Many **types and formats** of data from different sources. Makes integration, transformation, processing and storage hard. | Text, images, video, logs, JSON, XML |
| **Veracity** | **Quality / trustworthiness** of data. Data has **signal** (useful) and **noise** (useless). Data must be cleaned. | Online registration data = less noise; blog posts = more noise |
| **Value** | **Usefulness** of data to the business – the most important V. Raw data → information → cleaned → analysed. | Poor quality data can cause ~20% revenue loss |

> **Memory trick: V**ery **V**ast **V**aried, **V**erified, **V**aluable data = Volume, Velocity, Variety, Veracity, Value.

## 5. Challenges of Big Data (8)

| Challenge | Explanation |
|---|---|
| 1. Poor understanding of data | Professionals often don't know what the data is, where it came from, or what to do with it. |
| 2. Data storage issues | Data comes from different sources and formats, grows exponentially, needs lots of computing power and resources. |
| 3. Confusion in selecting a tool | Hard to choose between Cassandra, HBase, MapReduce etc. One wrong choice wastes money, time and effort. |
| 4. Data integration | Data from different sources may have no relationship between them, so combining is difficult; wrong integration gives inaccurate reports. |
| 5. Data security | Companies focus on storage/processing and leave security for later; open vulnerabilities attract hackers. |
| 6. Lack of data professionals | Big data is complex, tools keep upgrading, and professionals struggle to keep learning. |
| 7. Data quality | Mixed formats lower quality: irrelevant parts, inaccurate values, noise/outliers, missing points. |
| 8. Tolerance of unusual faults | An unexpected fault can destroy the whole process and force a restart from scratch; hard to tolerate because it needs intricate algorithms. |

## 6. Why Big Data? & How Does Big Data Work?

Big data alone cannot give business intelligence; it must be processed to give **actionable insights**. Three stages:

| 1. Data flow intake | 2. Data analysis | 3. Data-driven decisions |
|---|---|---|
| Huge, mixed data enters system. Cloud and distributed storage help. | System cleans and organizes data (like filing a pile of documents). Gives raw findings. | Interpret findings and create an evidence-based plan. |

## 7. Big Data vs Business Intelligence (BI)

| Point | Big Data | Business Intelligence |
|---|---|---|
| Data type | Large, complex, mostly unstructured | Structured data in data warehouses |
| Processing | Hadoop, Spark, NoSQL | BI tools: reports and visualizations |
| Storage | Distributed file system (HDFS) | Data warehouse |
| Techniques | MapReduce, machine learning | OLAP, data mining |
| Time | Real-time / near real-time | Historical data, not time-sensitive |
| Goal | Discover new insights and opportunities | Monitor performance, informed decisions |

**Analysis vs Reporting (syllabus):** *Reporting* = presenting data in organized form (tells **what** happened). *Analysis* = digging into data to find **why** and what to do next.

# 8. Big Data Architecture

**It must be able to:** capture data from different sources; clean and integrate different formats; sort and organize data; analyse data; find relationships and patterns; draw conclusions.

## 10 Layers (learn in order)

| Layer | What it does |
|---|---|
| 1. Data Sources | Absorbs and integrates data coming from various sources, at different speeds and formats. |
| 2. Ingestion | Validates, cleans, transforms, reduces and integrates data. **7 stages:** Identification, Filtration, Validation, Noise reduction, Transformation, Compression, Integration. |
| 3. Storage | **HDFS** stores huge data in **blocks**; fault tolerant, parallel; **write once read many**. NoSQL covers all storage needs. |
| 4. Physical Infrastructure | Hardware and network. Either a **cloud** or a distributed grid of **commodity** hardware. |
| 5. Platform Management | Tools and query languages to access NoSQL, uses HDFS. Building blocks: **MapReduce** (processing), **Hive** (data warehouse to read/write/manage), **Pig** (create apps that run MapReduce jobs), **HBase** (distributed scalable store), **ZooKeeper** (config info, naming, synchronization). |
| 6. Security | Authenticate nodes, file-layer encryption, key management service for trusted keys/certificates, validation tools, keep communication logs, secure communication using SSL. |
| 7. Monitoring | Monitoring systems aware of all configurations and functions of OS and hardware; machine communication; tools for data storage and visualization. |
| 8. Analytics Engine | Analyses huge unstructured data (text and statistical analytics). Types: **Search engine** (fast, iterative discovery) and **Real-time engine**. |
| 9. Visualization | Interpreting and viewing data from different perspectives, done by analysts/scientists. Works on aggregated data in ODS, data warehouse and marts (ODS gets data through Sqoop). |
| 10. Big Data Applications | **Horizontal** = solve problems common across industries. **Vertical** = solve an industry-specific problem. |

> **Order trick:** Sources → Ingestion → Storage → Infrastructure → Platform mgmt → Security → Monitoring → Analytics → Visualization → Applications.

## Big Data Technology Components

- Redundant physical infrastructure
- Security infrastructure
- Operational databases
- Organizing data services and tools
- Analytical data warehouses

## 9. Hadoop Environment and Tools

### MapReduce

- Designed by **Google** to run functions on large data in **batch mode**. A Java-based programming model for distributed computing.
- **Map:** splits the task across many systems, balances load and handles failures.
- **Reduce:** after computing, combines (aggregates) all results into the final answer.

### Big Table

- Developed by **Google**: distributed storage system for highly scalable structured data (tables with rows and columns).
- Different from relational model: a **sparse, distributed, persistent, multidimensional sorted map**. Stores huge data on commodity servers.

### Hadoop

- Apache software framework **derived from MapReduce and Big Table**. Runs MapReduce apps on large clusters of commodity hardware. (Yahoo! uses it.)
- Parallelizes processing across nodes to speed up computing and hide latency.
- **Two main components:** (1) scalable **distributed file system (HDFS)** supporting petabytes, (2) scalable **MapReduce engine** for batch results.

**Hadoop ecosystem (from diagram):** HDFS (storage), YARN (cluster resource management), MapReduce (processing), Pig (scripting), Hive (SQL query), HBase (columnar store), Mahout (machine learning), Sqoop & Flume (data collection), Oozie (workflow), ZooKeeper (coordination), Ambari (management/monitoring), Avro, Thrift, HCatalog.

## 10. Types of Big Data Analytics (4 types)

Going from simple to complex, value increases: **Descriptive → Diagnostic → Predictive → Prescriptive.**

| Type | Question | Techniques | Tools | Applications |
|---|---|---|---|---|
| **Descriptive** | What happened? | Data aggregation, data mining, visualization, statistical analysis | Excel, Tableau, Power BI, SQL | Business reporting (sales, KPIs), customer segmentation, market analysis, operational efficiency |
| **Diagnostic** | Why did it happen? | Correlation, regression, root cause analysis, data drilling, hypothesis testing | (BI/analysis tools) | HR, healthcare, manufacturing, IT, retail |
| **Predictive** | What could happen? | Regression, time series, machine learning, classification & clustering | R, Python (scikit-learn, TensorFlow), SAS, IBM SPSS | Risk management, customer retention (churn), sales forecasting, healthcare outbreaks |
| **Prescriptive** | What should we do? | Optimization, simulation, decision analysis, machine learning | Gurobi, IBM CPLEX, AnyLogic, MATLAB | Supply chain, revenue/pricing, personalized treatment, finance portfolios |

- **Descriptive** uses *historical* data to summarize the past; easy to visualize.
- **Diagnostic** explains the reason behind events and helps troubleshoot problems.
- **Predictive** uses historical data + statistical algorithms to forecast the future.
- **Prescriptive** goes beyond prediction: recommends the best action using data, algorithms and business rules.

## 11. Data Analytics Life Cycle (6 phases)

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| Discover | Data Preparation | Model Planning | Model Building | Communicate Results | Operationalize |

Simple meaning: understand the problem → collect and clean data → plan the model → build and test the model → share the results → put the solution into real use. (Slides only list the phase names; this line is a simple explanation.)

## 12. Last-Minute Revision

- **3 data types:** Structured (tables/SQL), Semi-structured (JSON/XML), Unstructured (media/text).
- **5 Vs:** Volume, Velocity, Variety, Veracity, Value (Value = most important).
- **HDFS:** blocks, commodity hardware, write once read many. **MapReduce:** Map (split) + Reduce (combine).
- **Hadoop = HDFS + MapReduce**, open source, based on Google MapReduce & Big Table.
- **Ingestion stages (7):** Identification, Filtration, Validation, Noise reduction, Transformation, Compression, Integration.
- **4 analytics:** What happened? Why? What could happen? What should we do?
- **Big Data vs BI:** unstructured + Hadoop + real-time vs structured + warehouse + historical.